

Security Surveillance System

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Abstract—This paper is based on making a smart Security surveillance system which uses the face recognition done with transferred learning using VGG-16, Siamese Network. This model takes the 8-10 image of the person as input value for training.

We are using transferred learning technique over VGG-16 to create 128D features. These features are then saved and used to compare with the features there are produced during the prediction of the person. If the person visiting your property is unknown then his/her image will be sent to the owner of the property using any media like mail(implemented in this project).

As we are using Raspberry Pi circuit thus we can provide a backup power so in case of power cut the security system can be still working.

Index Terms—component, formatting, style, styling, insert

I. INTRODUCTION

Recent world happenings have created a change in terms of security from "investigation" to "prevention of potentially catastrophic incidents". Existing digital video monitoring systems only provide the infrastructure for capturing, storing, and distributing video, leaving threat detection to human operators alone. Human monitoring of surveillance video is a very labour-intensive task. It's well acknowledged that viewing video feeds necessitates a higher degree of visual focus than most other activities. Specifically, caution, or the capacity to pay attention and respond to rare situations, is incredibly difficult and vulnerable to error due to attention lapses. So we have decided to try to build smart security surveillance system.

II. LITERATURE SURVEY

Many researches have been published in the area of face recognition and intruder detection in the recent years. Some of the papers are mentioned below and the main points of the paper:

A. Paper [1]

- Here the author primarily focusses on remote control techniques in Security Systems and Home Automation.
- They have used different gates for detection of intruder and fire in system
- Analyses different techniques of communication viz. Email Alerting, SMS Control, etc.
- Uses microcontroller, PC, Relay register for appliances, etc. Hardware.

B. Paper [2]

- Focuses on the Face Recognition enabled Internet based remote controlled Security Robot
- The face recognition approach possesses "invariant" recognition characteristics, including face recognition where facial expressions, viewing perspectives, three-dimensional poses, individual appearance, and lighting vary and occluding structure.
- Uses a Mechanical and Electronical Robot for security of Intruder detection

C. Paper [3]

- Focuses on the Face Recognition using Raspberry Pi
- The face recognition system implements "Boosted Cascade of Simple Features algorithm" to detect human faces. "Local Binary Pattern algorithm" to recognize these faces
- Uses a GPU enabled linux-based server for computation purposes.

D. Paper [4]

- Focuses on the integrating IoT and Face Recognition using Raspberry Pi.
- The face recognition system is implemented using Python library OpenCV, Python Language and a Raspberry Pi.
- It implements the intruder detection system using Raspberry pi camera module to take 30 pictures and detect faces using facial recognition algorithm.

E. Paper [5]

- Focuses on making a facial recognition system using opencv
- The face recognition system is implemented using Python library OpenCV, Python Language, a Raspberry Pi, Haar Cascade, Eigenfaces, Fisher faces, and a local binary Patterns Histogram
- It uses Haar Cascade to detect faces in an image and gives the co-ordinates of the face inside an image.
- After Face detection it is used to recognize Face.

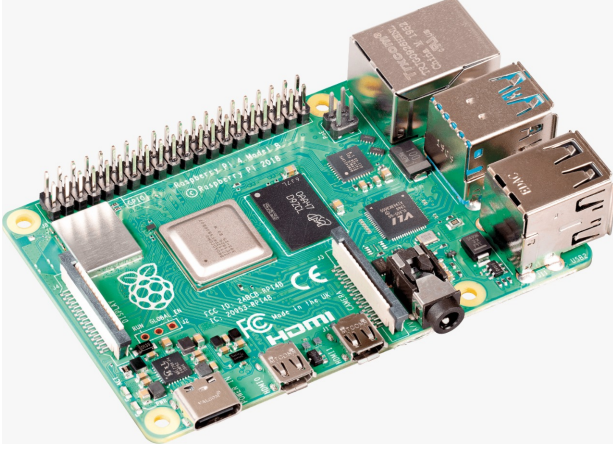
F. Paper [6]

- Uses vgg 16 architecture for face recognition
- Uses fully connected layers to fine tune vgg_16 model
- Uses OpenCV for Face detection

- Uses confusion matrix to evaluate performance
- Uses ROSE-Youtu Face Liveness Detection and some other faces as database for model

III. HARDWARE REQUIREMENTS

A. Raspberry Pi 4B+



The main project is run in a Raspberry Pi chip which itself can act as a computer with Raspberry Pi OS running in it. It works similar to the linux environment. The main advantage of this chip is that it is small as a credit card in size and has GPIO pins which helps to direct communicate with the sensors. It can also be connected to Monitor Keyboard and other peripherals with wired or wireless connections.

B. HC-SR04 Ultrasonic Sensor



The ultrasonic sensor is used to measure the distance between the sensor and the object kept in the way of it. It transmits ultrasonic wave, when this wave gets objected by some material, it comes back to the sensor. This phenomenon helps

to get the distance with the help of time and speed of the wave

$$Distance = (Speed * time)/2 \quad (1)$$

Therefore

$$Distance = (34300 * time)/2 \quad (2)$$

We need to divide the distance by 2 as it is calculated twice; once for going towards object and second time for coming towards the object. It has four pins Vcc, Trig, Echo, Gnd. Vcc and gnd are the input voltage(5V) and the ground pin. The trigger is used to activate the wave and the echo is used to read the received wave.

C. Raspberry Pi Camera Module

A raspberry pi camera module is a low cost HD camera built for Raspberry pi implementations. It is available for both wired and wireless connections which can directly be connected to the Pi device. This will be used to give system the image input for the face recognition.

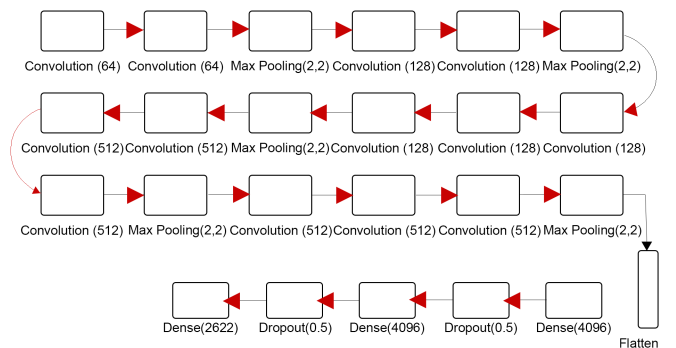
IV. SOFTWARE REQUIREMENTS

- Python3
- OpenCV
- Dlib
- Tensorflow
- Anaconda
- Pickle
- Numpy
- Imutils
- RPi.GPIO(to get the readings form ultrasonic sensor)
- SMTP lib (to send the mail to the user)

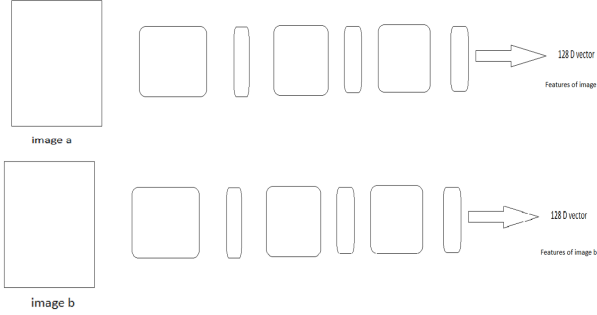
V. METHODOLOGIES

A. Training Model

1) *Initial Training:* For the initial training of the network we are using the VGG_face Dataset. This dataset consists of images 2622 of different celebrities.

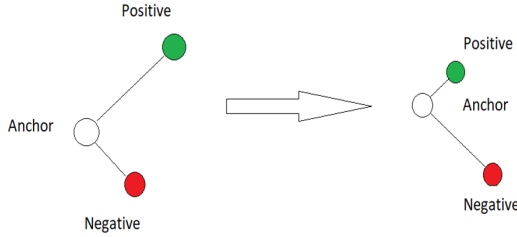


2) *Training of faces to be detected:* After the last dense layer in the model we add a pop and a dense layer of 128 neurons. We then train the model with the help of input images(8-10) and the weights of the previous model by applying gradient decent on the loss function (triplet loss).



B. Triplet Loss

Triplet loss is a function which takes 3 vectors as a input. One is the vector of the person which we want to predict(positive). Another one is the vector of the same person(anchor). And the third one is the image of the random person from the dataset VGG_face(negative). It finds the distance between anchor and positive(lets assume x) and, anchor and negative(lets assume y). Then the comparison is done between the x and y.



And the third one is the image of the random person from the dataset VGG_face(negative). It finds the distance between anchor and positive(lets assume x) and, anchor and negative(lets assume y). Then the comparison is done between the x and y.

$$X = (\text{anchor} - \text{positive})^2 \quad (3)$$

$$Y = (\text{anchor} - \text{negative})^2 \quad (4)$$

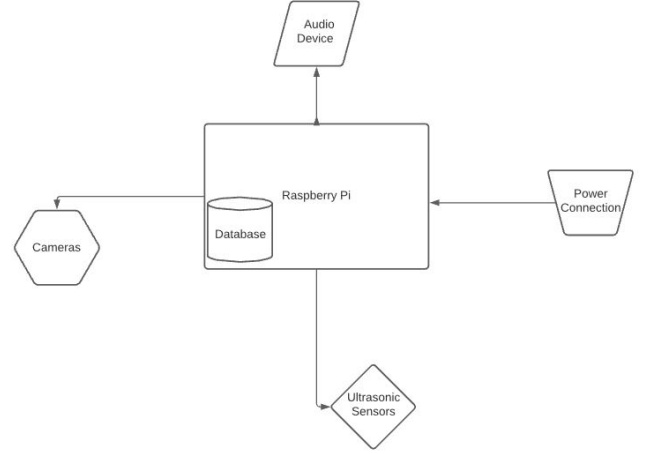
$$X + \alpha \leq Y \quad (5)$$

Here alpha is the hyperparameter which decides the min distance between X and Y And the gradient decent is applied thus the vector of the anchor and positive comes closer.

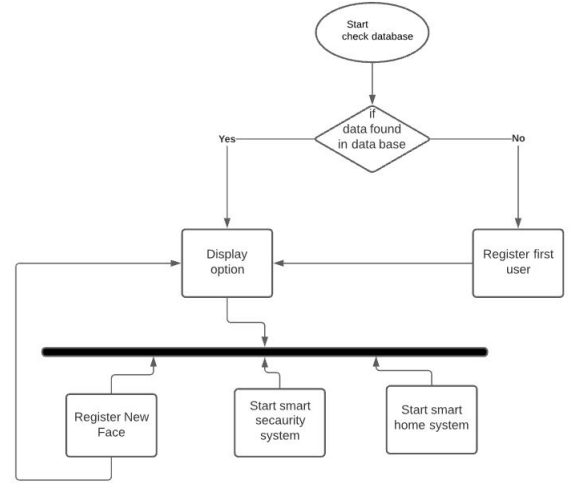
C. Prediction

At the time of prediction we generate the features in the same way and then compare it with the features obtained at the time of the registration as these features must be almost same (80

VI. IMPLEMENTATION



The database is initially empty. An Ultrasonic Sensor is set near the possible inlet (door). A camera is set in such a way that the full view of the door is visible. The cameras and the sensors are then connected to the Raspberry Pi using GPIO pins and standard connection for camera respectively. And the system is then provided by the adequate power supply.



When the security system starts it first checks the local database. If the database is empty it gets the first face from the user. The pic from the camera is taken and checked with the dlib face detector for the presence of the face. If the face is present then it is provided to the network and if not then the message is displayed for the user to come in front of camera.

Using the network, the face encoding of the respective person is extracted. The face recognition program takes 8-10 image as input and outputs 128D feature vector. The Encoding obtained is then stored in the pickle format.

Now after the installation of the first face the ultrasonic sensor starts detecting the distance. With the help of time

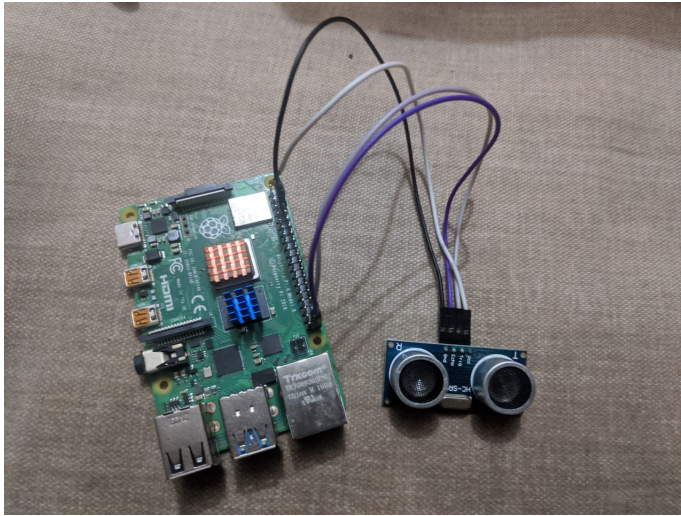
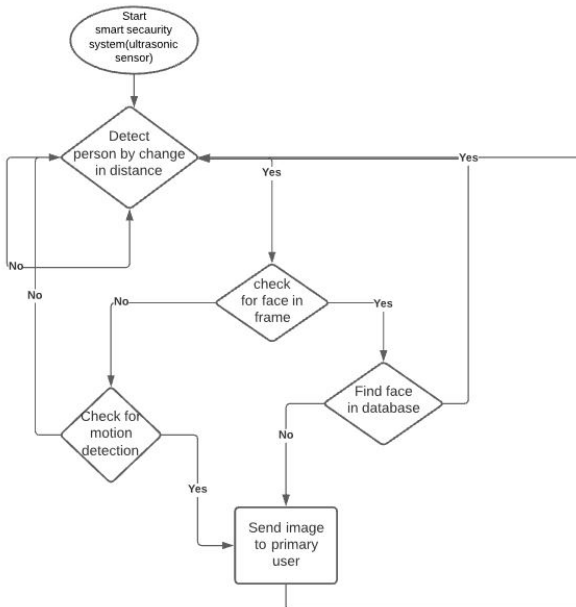


Fig. 1. configuration of ultrasonic sensor to Raspberry pi
Vcc(ultrasonic sensor) to pin4 5v output
Ground(ultrasonic sensor) to pin6 (ground)
Trig(ultrasonic sensor) to pin38 (GPIO 20)
Echo(ultrasonic sensor) to pin40 (GPIO 21)

module and GPIO output and input, the sending and the receiving time of the waves can be recorded.

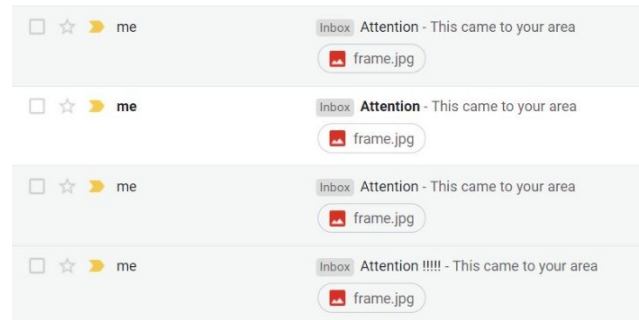
This time is then multiplied with the speed of the ultrasonic waves i.e. 34300. To obtain the actual distance we need to divide the distance by 2 as the distance obtained by this steps is the distance for the waves that goes to the object and comes back again to the sensor. We need the distance for only one way.



The distance is calculated in a loop until any object comes in between. If the person moves through the door, distance is the ultrasonic sensor gets fluctuated. As the distance is fluctuated the face prediction is method is called.

Hear the method gets the image and pass it to the face recognition prediction function. This basically gets the 128D vectors of the face entered and checks with that are already present with us. The features are brought back from the pickle and converted into a vector and provided to the algorithm to compare. If the face matches the loop resumes without any action and if the face does not match the picture for the face is then sent to the mail method as an argument.

The mail method then gets the email of the primary person from the data base. Now using the smtplib it sends the mail to the person using the mail id which was given at the time of face registration.



This mail is sent with the Subject “Attention !!!!!”, and the message is ”This came to your area,” with a photo of the claim added as an addition to the mail. Thus as a result the person will get the notification that any unauthorised person has entered the premises.

VII. RESULTS AND DISCUSSIONS

The smart security surveillance system results in the notification via email when an unauthorised person enters your property using face data. This system enables the person to rely on it in the case of power cut to as the system can work on GPIO power input using small power source.

As this project is based on Raspberry Pi it can also be extended to the Smart Home System with a help of Node MCU which can control various home appliances like lights, fans, etc.

This project can also be extend in a way by giving an option to the user to live stream the video when the notification of interference of any unauthorised person is there by the usage of the server to stream the video.

VIII. CONCLUSION

There are some systems already successful in the surveillance using the man power monitoring the cameras 24*7. This project will do the surveillance removing the 24*7 monitoring of human to the camera but will itself send the notification and hence will make the work one step easy.

REFERENCES

- [1] K. Balasubramanian and A. Cellatoglu on "Analysis of Remote Control Techniques Employed in Home Automation and Security Systems".
- [2] James N. K. Liu, Member, IEEE, Meng Wang, Student Member, IEEE, and Bo Feng, Student Member, IEEE on "iBotGuard: An Internet-Based Intelligent Robot Security System Using Invariant Face Recognition Against Intruder".
- [3] Ayman A. Wazwaz, Amir O. Herbawi, Mohammad J. Teeti and Sajed Y. Hmeed on "Raspberry Pi and Computers-Based Face Detection and Recognition System".
- [4] Ibrahim Mohammad Sayem and Mohammad Sanaullah Chowdhury on "Integrating Face Recognition Security System With the Internet of Things".
- [5] Maliha Khan, Sudeshna Chakraborty, Rani Astya and Shaveta Khepra on "Face Detection and Recognition Using OpenCV".
- [6] Anugrah Bintang Perdana, Adhi Prahara on "Face Recognition Using Light-Convolutional Neural Networks Based on Modified VGG16 Model".